STA 408 Lec#01

Design and Analysis of experiment

Planning an experiment to obtain appropriate data and drawing inference out of the data with respect to any problem under Investigation is know as Design & analysis of experiment.

Why all this ?

- Experiment :-

An experiment deliberately imposes a treatment on a group of objects or subjects in the Interest of observing the response.

- Treatments :- *Quiz

In experiments, a treatment is something that researchers administer to experimental units.

For Example, a Corn field is divided into four, each part is 'treated' with a different fertilizer to see which produces the most corn.

- FactoR :- * Quiz

A factor of an experiment is controlled Independent Variable; a Variable whose levels are set by the experimenter.

A factor is a general type * Quiz or category of treatments. Different treatments constitute different levels of a factor.

* Quiz

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Example:

Factor	Vitamin C		
	0 mg	250 mg	500 mg
Vitamin E			
0 mg	T_1	T_2	T_3
400 mg	T_4	T_5	T_6

- **Experimental unit** - Quiz

The experimental unit is the basic entity or Unit on which the experiment is performed (Quiz) i.e. it is an object to which the treatment is applied & in which the variable under Investigation is measured & Analyzed.

Identify The Experimental unit ?

- A teacher practice the different teaching methods on different groups in her class to see which yields the best results.
- A doctor treats a patient with a skin condition with different creams to see which is most effective.

- **Experimental ERROR**

It Describes the variation among Identically and Independently treated experimental units.

The various origin of experimental error Include

1. The natural variation among the experimental units

Inability to reproduce the treatment conditions exactly from one unit to another - Quiz
Interaction of treatments and experimental units.
Any other extraneous factor that Influence the measured characteristics.

Types of ERRORS

- Systematic errors
- Random errors

Experimental Design

An experimental Design is a plan to collect the data relevent to the problem under Investigation. In such a way as to provide a basis for valid and objective Inferences about the standard problem. The plan usually consist of the selection of the treatments, Specifications of the experimental layouts, allocation of the treatments and collection of observation for analysis.

Pupose of Experimental Design

1. Get maximum Information for minimum expenditure in the minimum possible time
2. Help to reduce the experimental error
3. To ignore Quiz Spurious effects, if any.
4. To evaluate and examine the outcomes Critically and logically.

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Steps In Designing An experiment

A statistically designed experiment may consist of the following steps.

1. Statement the problem
2. Formulation of hypothesis
3. Choice of response or dependent Variable.
4. Selection of the Factors to be Varied.
5. Choice of level of these factors i.e qualitative or quantitative and fixed or random.
6. How Factors are to be combined.
7. Number of observations to be taken.
8. Methods of randomization to be used.
9. Mathematical model to describe the experiment.

Basic Principles of experimental Design

The basic principles of experimental designs are:

1. Randomization
2. Replication
3. Local Control

These principles make a valid test of significance Possible.

Randomization:

It is a random process of assigning treatments to the experimental units.

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Randomization is usually done by using tables of random numbers, by drawing numbered card from a well shuffled pack of cards or by drawing numbered balls from a well-shaken container.

Purpose of Randomization:

* Quiz

1. To remove bias and other source of extraneous variation, which are not controllable.
2. To form the basis of any valid statistical test.

* Quiz

Replication:-

Quiz

Replication is basically a repetition of the basic experiment.

In other words, it is a complete run for all the treatments to be tested in the experiment.

In all experiments, some variation is introduced because of the fact that the experimental units such as individuals or plots of land in agriculture experiments, cannot be physically identical. This type of variation can be removed by using a number of experimental units. We therefore perform the experiment more than once, i.e., we repeat the basic experiment. An individual repetition is called a replicate. Quiz*

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Replication is used ?

1. To secure more accurate estimate of the experimental error. Quiz
2. To decrease the experimental error and thereby to Increase precision Quiz
3. To obtain more precise estimate of the mean effect of a treatment - Quiz

3 Local Control :- Quiz

A term referring to the amount of balancing, blocking and grouping of the experimental units. Balancing means that treatments should be assigned to the experimental units in such a way that the result is a balanced arrangements of the treatment Blocking means that like experimental units should be collected together to form a relatively Quiz homogeneous group. A block is also a replicate. Quiz The main purpose of the principle of local control is to Increase the efficiency of an experimental design by decreasing the experimental error.

TYPES of DESIGNS

1 Systematic Design
2 Randomized Design

} * Quiz

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Systematic Design:-

In systematic designs treatments are applied to the experimental units by some systematic pattern, i.e by the choice of the experimenter. Quiz

For example:-

The experimenter wishes to test three treatments and he decides to have four repetitions of each treatment. The arrangements of three treatments over the experimental area could be

Randomized Design:-

Quiz

In randomized Designs, as the treatments are applied randomly therefore the conclusion drawn are supported by statistical test.

In this way, Inference are applicable in wider/large range and the random process minimizes the systematic error. Quiz

The analysis of variance techniques are also suitable to randomized design only. Quiz

Classifications of Randomized Designs

Completely Randomized Design (CRD)

Randomized Complete Block Design (RCBD)

Latin Square Design. Quiz*

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EXAMPLE

A market gardner wants to test three types of Peas, A, B, C on his land. He has a square plot which he divides into nine equal squares, three to be planted with each type of Pea. The problem which he then faces is which Squares to Plant with which type.

1st method is CRD (completely randomized Design)

C	A	C
B	A	A
B	B	C

See handout

Lec # 02 STA 408

Contents

1. Basic Concepts of ANOVA
2. CR Design

Analysis of Variance: A conceptual overview

Analysis of variance (ANOVA) can be used to test for the equality of three or more Population means. Quiz

Data obtained from observational or

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experimental studies can be used for the analysis we want to use the sample results to test the following hypothesis:

H_0: $\mu_1 = \mu_2 = \mu_3 = \cdots \mu_k$

H_a: Not all population means are equal

H_0: $H_1 = H_2$

H_A: $H_1 \neq H_2$

Quiz

- If H_0 is rejected, we cannot conclude that all population means are different.
- Rejected H_0 means that at least two population means have different values.

Assumptions for Analysis of Variance

Quiz

- For each population, the response (dependent variable) is normally distributed.
- The variance of the response variable, denoted σ^2, is the same for all of the populations.
- The observations must be independent.

Quiz

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Sampling Distribution of $\bar{x}$ Given H_0 is True

$$\mu_1 = \mu_2 = \mu_3 = \boxed{\mu}$$

→ Sample means are close together because there is only one Sampling Distribution when H_0 is true.

$$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$$

Sampling Distribution of $\bar{x}$ Given H_0 is False

→ Sample means come from different Sampling Distributions and are not close together when H_0 is False.

Analysis of Variance

1. Between-Treatments Estimate of Population Variance
2. Within Treatments Estimate of Population Variance
3. Comparing the Variance Estimates: The F Test Quiz
4. ANOVA Table

B/W Treatments Estimate of Population Varian

→ The estimate of σ^2 based on the variation of the Sample means is called the mean Square due to treatments and Quiz

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is denoted by MSTR (mean square Treatment):

$$MSTR = \frac{\sum_{j=1}^{k} n_j (\bar{x}_j - \bar{\bar{x}})^2}{k-1} \longrightarrow SSTR \text{ (sum of squares due to error)}$$

$k-1 \rightarrow$ D.f

$$SSTR = \sum_{j=1}^{k} n_j (\bar{x}_j - \bar{\bar{x}})^2$$

Here $k-1$ is the Degree of freedom associated with SSTR Quiz

Comparing the Variance Estimate: The F test

→ if the null hypothesis is true and the ANOVA assumptions are valid, the Sampling Distribution of MSTR/MSE is an F Distribution with MSTR d.f equal to $k-1$ & MSE d.f. equal to $n_T - k$ Quiz.

→ if the means of the k populations are not equal, the value of MSTR/MSE will be inflated bez MSTR overestimates σ^2.

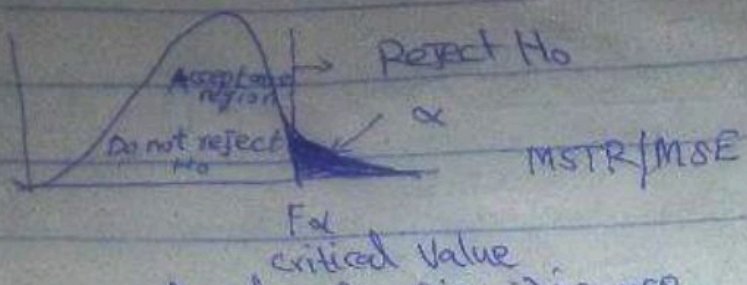
→ Hence, we will reject H_0 if the resulting value of MSTR/MSE Quiz apears to be too large to have been selected at random from the appropriate F Distribution.

$$F = \frac{MSTR}{MSE} \sim F(k-1, n-k)$$

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Sampling Distribution of (MSTR/MSE)



α = level of significance

Test for the Equality of k Population Means

1. Hypothesis

$H_0 : \mu_1 = \mu_2 = \mu_3 = \cdots \mu_k$

H_a: Not all population means are equal

2. Test Statistics :-

$$F = \frac{MSTR}{MSE}$$ Quiz...

3. ~~Reaction~~ Rejection Rule

→ P-value Approach :⇒ Reject H_0 if P-value ≤ α

→ Critical value Approach :⇒ Reject H_0 if $F \geq F_\alpha$

Where the value of F_α is based on an F-distribution with k-1 numerator d.f and $n_T - k$ denominator d.f.

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ANOVA Table

The Anova Table decomposes the variance into two Components (Sources): a b/w-group component (treatments) and a within-group Component (error). The f- ratio, which in this case equal 15.6234, is a ratio of the b/w-group estimate to the within-group estimate. Since the p-value of the F-Test is less than 0.05, there is a statistically significance difference b/w the means from one level to another at the 95.0% confidence level.

ANOVA Table For hours by brand

Source	Sum of Sq (SS)	Df	Mean Sq (MS)	F Ratio	P-value
Between groups	1.08917	3	0.363057	15.62	0.0000
within-groups	0.557714	24	0.0232381		
Total (corr.)	1.64689	27			

Completely Randomized Design (CRD)

A completely randomized (CR) design, which is the simplest type of the basic designs, may be defined as a design in which the treatments are assigned to experimental unit completely at random, that is the randomization is done without any restrictions. The design is

Quiz

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Completely flexible, i.e. any number of treatments and any number of units per treatment may be used. Moreover, the number of units per treatment need to be equal.

A completely randomized design is considered to be most useful in situations where (i) the experimental units are homogeneous *Quiz, (ii) the experiments are small such as laboratory experiments and (iii) some experimental units are likely to be destroyed or fail to respond *Quiz

→ when the samples of experimental units for each treatment are random and Independent of each other. Quiz

→ Design is used to compare the treatment means;

$H_0 : \mu_1 = \mu_2 = \mu_3 \cdots = \mu_k$

H_a: At least two of the treatment mean differ

- The hypothesis are tested by comparing the differences b/w the treatment means.
- Test Statistics is calculated using measure of Variability within treatment groups and measure of variability b/w treatment groups *Quiz

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Steps for conducting an Analysis of variance (ANOVA) for a completely Randomized Design

1 Assure randomness of design, and Independence, randomness of Sample.
2 Check normality, equal variance assumptions
3 Create ANOVA Summary table
4 conduct multiple comparisons for pairs of means as necessary / desired

Assumptions

1- Normality :-

You can check on normality using

1- Normal probability plot
2- Kolmogorve test

2- Constant Variance:-

You can check on homogeneity of variances using

1- Histogram plot
2- leven's Test.

CRD

ANOVA Summary Table for a completely Randomize desig

Source	df	SS	MS	F
Treatments	k-1	SST	$MST = \frac{SST}{k-1}$	$\frac{MST}{MSE}$
Error	n-k	SSE	$MSE = \frac{SSE}{n-k}$	
Total	n-1	SS(Total)		

χ^2

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Advantages of a CRD

a Very flexible design (i.e. number of treatments and replicates is only by the available number of experimental units).

b Statistical analysis is simple compared to other designs.

c Loss of Information due to missing data is small compared to other designs due to the larger number of degree of freedom for the error Source of variation.

Disadvantages:-

a If experimental units are not homogeneous and you fail to minimize this variation using blocking, there may be a loss of precision.

b Usually the least efficient design unless experimental units are homogeneous. Quiz

c Not suited for a large number of treatments.

Experimental Layout:-

An example of the experimental layout for a completely randomized design (CR) using four treatments A, B, C and D, each repeated 3 times, is given below:

A	B	C	D
B	C	A	D
D	A	A	B

r = 3

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Observations	Treatments 1	2	3	...	j	...	k
1	Y_{11}	Y_{12}	Y_{13}	...	Y_{1j}	...	Y_{1k}
2	Y_{21}	Y_{22}	Y_{23}	...	Y_{2j}	...	Y_{2k}
3	Y_{31}	Y_{32}	Y_{33}	...	Y_{3j}	...	Y_{3k}
⋮	⋮	⋮	⋮		⋮		⋮
i							
⋮	⋮	⋮	⋮		⋮		⋮
r_j	$Y_{r_j 1}$	$Y_{r_j 2}$	...				$Y_{r_j k}$
Total	T_1	T_2	T_3	...			$T_{\cdot k}$
Mean	$\bar{Y}_{\cdot 1}$	$\bar{Y}_{\cdot 2}$					$\bar{Y}_{\cdot k}$

Total SS = Treatment SS + Error SS

$$\sum_{j=1}^{k}\sum_{i=1}^{r_j}(Y_{ij}-\bar{Y}_{\cdot\cdot})^2 = \sum_{j=1}^{k} r_j(\bar{Y}_{\cdot j}-\bar{Y}_{\cdot\cdot})^2 + \sum_{j=1}^{k}\sum_{i=1}^{r_j}(Y_{ij}-\bar{Y}_{\cdot j})^2$$

Df = $(n-1)$ $\quad$ $(k-1)$ $\quad$ $(n-k)$

D.f

Total = $(n-1)$

Treatment = $(k-1)$

Error = $(n-k)$

$$F = \frac{MSTR}{MSE}$$

$$F = \frac{\text{MS b/w treatment}}{\text{MS within treatment}}$$

$$F \sim f_{\alpha}(k-1, n-k)$$

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Variety

A	B	C
23	18	16
26	28	25
20	17	12
17	21	14

Test the hypothesis that the three varieties of potatoes are not different in yielding capabilities.

Hypothesis Testing procedure

i $H_0: \mu_A = \mu_B = \mu_C$

H_A: Not all the three means are equal

ii Level of significance:

$\alpha = 0.05$

iii Test-statistics

$$F = \frac{MST}{MSE}$$

which if H_0 is true, has an F-distribution with

$\nu_1 = k-1 = 3-1 = 2$

$\nu_2 = n-k = 12-3 = 9$

degree of freedom

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Anova Table in the case of the completely Randomized CR Design

Source of variation	df	Sum of sq	Mean Sq	F
Between treatments	k-1	SST	MST	MST/MSE
within treatments	n-k	SSE	MSE	----
Total	n-1	Tss		----

A	B	C	Total	$\sum_i X_{ij}^2$
23 (529)	18 (324)	16 (256)		1109 ⇒ column wise
26 (676)	28 (784)	25 (625)		2085
20 (400)	17 (289)	12 (144)		833
17 (289)	21 (441)	14 (196)		926
$T_{.j}$ 86	84	67	add ⇒ 237	(4953) ↑ Check
$T_{.j}^2$ 7396	7056	4489	18941	←
$\sum_i X_{ij}^2$ 1894 ⇓ Row wise	1838	1221	4953	

$$\sum_i \sum_j X_{ij}^2 = 4953$$

$$T_{..} = \sum T_{.j} = 237$$

We have

$$C.F = \frac{T_{..}^2}{n} = \frac{(237)^2}{12} = 4680.75$$

Hence, the total sum of sq is given

$$TSS = \sum_i \sum_j X_{ij}^2 - C.F$$

$$= 4953 - 4680.75$$

$$= 272.5$$

Also, we have

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$$SS\,Treatment = SST = \frac{\sum T_j^2}{r} - C.F$$

$$= \frac{18941}{4} - 4680.75$$

$$SST = 54.50$$

And, hence:-

$$SS\,Error = SSE = TSS - SST$$

$$= 272.25 - 54.50$$

$$SSE = 217.75$$

$$'MS\,Treatment' = \frac{SST}{d.f} \qquad n-1$$

$$= \frac{54.50}{3-1} = \frac{54.50}{2} = 27.25$$

$$MSE = \frac{SSE}{d.f} = \frac{217.75}{9} \qquad n-k = 12-3 = 9$$

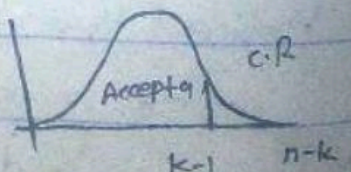
$$MSE = 24.19$$

ANOVA Table

Source variation	d.f	Sum of sq	mean sq	Comp F
B/w Treatment	2	54.50	27.25	$\frac{27.25}{24.19} = 1.13$
Error	9	217.75	24.19	
Total	11	272.25		

Critical Region

$$F \geq F_{\alpha}(k-1, n-k)$$



Df (2, 9)

Since

$$\mu_1 = k-1 = 3-1 = 2$$

$$\mu_2 = n-k = 12-3 = 9 \quad d.f$$

Calculated = 1.13

$$F \geq f_{0.05}(2, 9) = 4.26$$

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vi) Conclusion :-

Since the computed value of $F = 1.13$ doesn't fall in the CR, so we accept our null hypothesis and may conclude that, on the average, there is no difference among the yielding capabilities of the three varieties of potatoes.

EXAMPLE :-

Three Fertilizer treatments A, B, C each applied to seven Plots of ~~Arbi~~ Strawberry plants, resulted in the following wieghts of crops (lb/Plot):

A	B	C
24	46	32
18	39	30
18	37	26
20	50	41
27	44	36
17	45	28
15	30	27

Perform the analysis of variance to test the hypothesis of no difference in the treatment effects - use 0.05 level of significance.

Sol :-

1. Formulation of Hypothesis :-

$H_0 : \mu_1 = \mu_2 = \mu_3$

H_1 : At least two mean are not equal

2. Level of Significance

$\alpha = 0.05$

3. Test - Statistics :

$$F = \frac{MST}{MSE}$$

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Calculation:

							T.j
A	24	18	18	29	22	17	15	143
B	46	39	37	50	44	45	30	291
C	32	30	26	41	36	28	27	220
Total								654 (T..)

Here, $T_1 = 143$

$T_2 = 291$

$T_3 = 220$

$T = 654$

$TSS = \sum\sum X_{ij}^2 - CF$ $\qquad CF = \frac{(T..)^2}{n}$

$CF = \frac{T..^2}{N}$, where $N = nk = 7(3) = 21$

$CF = \frac{(654)^2}{21} = 20367.43$

Total SS $= \sum\sum X_{ij}^2 - CF$

$TSS = 24^2 + 18^2 + 18^2 + 29^2 + 22^2 + 17^2 + 15^2 - 20367.43$

$= 22520 - 20367.43$

$= 2152.27$

$TreatSS = \frac{\sum_{j=1} T.j^2}{n} - CF$

$= \frac{1}{7}\left[(143)^2 + (291)^2 + (220)^2\right] - 20367.43$

$= 21932.86 - 20367.43$

$= 1564.43$

Error SS = Total SS − Treatment SS

[redacted]

$21525.7 - 1564.43 = 587.14$

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Source of variation	Sum of Sq	D.f	Mean Sq	F ratio
Treatment	1565.43	k-1=2	=1564.43/2 = 782.72	F=782.72/32.62
Error	587.14	N-k=18	=587.14/18	=24.00
Total	2152.57	N-1=20	= 32.62	

Critical Region :-

Reject H_0, if

$F_{cal} \geq F_{\alpha(1-1, N-k)}$

$24.00 \geq F_{0.05}(2, 18)$

$24.00 > 3.55$

We reject H_0

Conclusion:

Since the value 2400 > 3.55 fall in the critical Region, so we reject H_0 or ~~false~~ also we conclude that the there is difference b/w the treatment effects.

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Lecture # 03

STA 408

Randomized Complete Block Design

Testing the effects of two Independent variables or factor on one dependent variable.

Solving Hypothesis-Testing Problems

Step 1: State the hypothesis & Identify the claim.

Step 2: Find the critical value(s) from the appropriate table.

Step 3: Compute the test value (ANOVA Table)

Step 4: Make the decision to reject or not reject the null hypothesis.

Step 5: Summarize the results.

Main Points of RCBD :-

- Experimental meterial is divided into groups or blocks Quiz
- Each block contains a complete set of treatments Quiz
- Treatments are assigned randomly
  - Randomization is restricted within block Quiz
- A new randomization is done for each block/group. Quiz

It has different layouts: 2x3, 3x5 etc

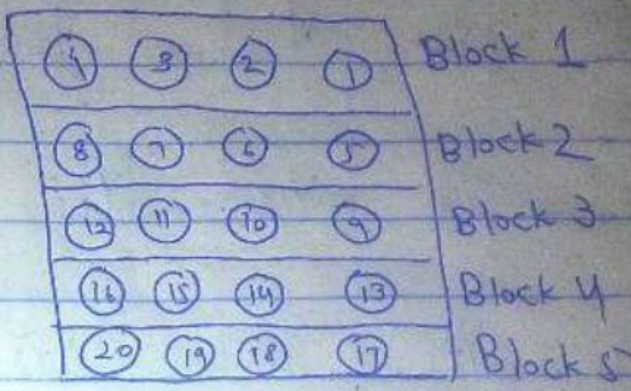
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- The RCB is perhaps the most frequently used experimental design

CRD Vs RCBD





Experimental Layout

Allot treatment randomly

A B C D

Block-I	B	C	A	D
Block II	A	D	B	C
Block III	C	A	B	D

Statistical model

- The linear model for this design is:

$$Y_{ij} = \mu + B_i + T_j + e_{ij}$$

- where;

μ = True mean effect

T = Effect of treatment

B = Effect of block/group

e = effect of random error

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Data Computations Table

Block (Groups)	Treatments 1	2	...	j	...	k	Total	Mean
1	Y_{11}	Y_{12}	...	Y_{1j}	...	Y_{1k}	$B_{1\cdot}$	$\bar{Y}_{1\cdot}$
2	Y_{21}	Y_{22}	...	Y_{2j}	...	Y_{2k}	$B_{2\cdot}$	$\bar{Y}_{2\cdot}$
⋮	⋮							
i	Y_{i1}	Y_{i2}	...	Y_{ij}		Y_{ik}	$B_{i\cdot}$	
r	Y_{r1}	Y_{r2}	...	Y_{rj}		Y_{rk}	$B_{r\cdot}$	
Total	$T_{\cdot 1}$	$T_{\cdot 2}$		$T_{\cdot j}$		$T_{\cdot k}$		
Means	$\bar{Y}_{\cdot 1}$							

Block

Partitioning of Variations

[Total Variation] = [variation due to treatment] + [Variation due to Block/Group] + [Variation due to unknown factor]

Total SS = Treatment SS + Block SS + Error SS

$$\sum_i \sum_j (Y_{ij} - \bar{Y})^2 = k\sum_j (\bar{Y}_{\cdot j} - \bar{Y}_{\cdot\cdot})^2 + k\sum_i (\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot})^2 + \sum_i \sum_j (Y_{ij} - \bar{Y}_{\cdot j} - \bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot})$$

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ANOVA ~~For~~ Table for RCB

Source of Variation	Degree of freedom	SS = Sum of square	MS = Mean Squarey	F-test
Variations due to Treatment	$k-1$	$SSTr = r\sum_j(\bar{Y}_{.j} - \bar{Y}_{..})^2$	$S_t^2 = \frac{SST_r}{k-1}$	$F_t = \frac{S_t^2}{S_e^2} = \frac{MST}{MSE}$
Variations due to Blocks	$r-1$	$SSB = k\sum_i(\bar{Y}_{i.} - \bar{Y}_{..})^2$	$S_b^2 = \frac{SSB}{r-1}$	$F_b = \frac{S_b^2}{S_e^2} = \frac{MSB}{MSE}$
Variations due to ~~Error~~	$(k-1)(r-1)$	SSE = By subtraction	$S_e^2 = \frac{SSE}{(r-1)(k-1)}$	
Total	$rk-1$	$SST = \sum_i\sum_j(Y_{ij} - \bar{Y}_{..})^2$	----	

EXAMPLE

- Four Verities of wheat are tested in randomize Complete Clock design in Four replications / blocks. Yield in kg / Plot is shown in the table below.
- Test the hypothesis that there is no difference in the means of four varieties.

Blocks	Varieties of Wheat			
	V_1	V_2	V_3	V_4
I	2	5	4	1
II	2	3	3	1
III	4	6	6	2
IV	1	4	2	3

Solution:

Testing Hypothesis

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Step 1: State the hypothesis

$H_0: T_j = 0$ (There is no difference in the mean of four varieties of wheat)

$H_1: T_j \neq 0$ Not all the four means are equal

$H_0: B_i = 0$

$H_0: B_i \neq 0$

Step 2:

We set the level of significance alpha

$\alpha = 0.05$ (5%)

Step 3:

The test-statistics formula to be used is:

$$F = \frac{S_t^2}{S_e^2} = \frac{MS(\text{Treatment})}{MS(\text{Error})}$$

It follows F-distribution with degree of freedom

df $= V_1 =$ from treatment $= k-1$

df $V_2 =$ from error $= (k-1)(r-1)$

Blocks Varieties Treatment

	V1	V2	V3	V4	Block total	(Block Tot)^2	Sum of Sq of each row
I	2	5	4	1	2+5+4+1=12	$12^2=144$	$2^2+5^2+4^2+1^2=46$
II	2	3	3	1	9	81	23
III	4	6	6	2	18	324	92
IV	1	4	2	3	10	100	30
Varieties Total	9	18	15	7	⇒ 49		
(Varieties Total)^2	81	324	225	49			
Sum of square of each value	25	86	65	15			⇒ 191

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Date: 1/20

Step1: State the hypothesis

$H_0: T_j = 0$ (There is no difference is the mean of four varieties of wheat)

$H_1: T_j \neq 0$ Not all the four means are equal

$H_0: B_i = 0$

$H_0: B_i \neq 0$

Step2:

we set the level of significance alpha

$\alpha = 0.05$ (5%)

Step3:

The test-Statistics formula to be used is:

$$F = \frac{S_t^2}{S_e^2} = \frac{MS(Treatment)}{MS(Error)}$$

it follows F-distribution with degree of freedom

$df = V_1 =$ from treatment $= k-1$

$df \quad V_2 =$ from error $= (k-1)(r-1)$

Blocks varieties Treatment

	V1	V2	V3	V4	Block total	(Block tot)^2	Sum of Sq of each value
I	2	5	4	1	2+5+4+1=12	$12^2=144$	$2^2+5^2+4^2+1^2=46$
II	2	3	3	1	9	81	23
III	4	6	6	2	18	324	92
IV	1	4	2	3	10	100	30
Varieties Total	9	18	15	7	⇒ 49		
(Varieties Total)^2	81	324	225	49			
Sum of square of each value	25	86	65	15			→ 191

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STA 408 Mathematics

Lecture # 04

RCBD with replications within Blocks:

Let us assume that each treatment is replicated r times within each block, So each block contain rk plots. The experiment becomes analogous to two way classifications with each cell containing r observations. This leads to new Statistical model in the form

$$Y_{ijl} = \mu + B_i + \tau_j + (B\tau)_{ij} + e_{ijl}$$

where;

$i = 1, 2, \ldots r$ It shows for blocks/rows

$j = 1, 2 \ldots k$ " " " no of Treatment

$l = 1, 2 \ldots n$ " " no of Replications

Missing observations in RCBD

Sometimes it So happens that Some experimental units or observations are missing.

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Suppose,

Blocks	Treatments 1	2	...	k	Block Total
1	x	Y_{12}	...	Y_{1k}	$B_1' + x$
2	Y_{21}	Y_{22}	...	Y_{2k}	B_2
⋮	⋮	⋮		⋮	⋮
r	Y_{r1}	Y_{r2}	...	Y_{rk}	B_r
Treatment Total	$T_{.1}' + x$	$T_{.2}$	...	$T_{.k}$	$T_{..}' + x$

Performing the analysis of variance, we get the following Sum of Squares;

Source of Variation	Sum of Squares
Treatments	$\frac{(T_{.1}+x)^2}{r} + \frac{T_{.2}^2}{r} + \cdots + \frac{T_{.1}^2}{r} - \frac{(T_{..}'+x)^2}{rk}$
Blocks	$\frac{(B_1'+x)^2}{k} + \frac{B_2^2}{k} + \cdots + \frac{B_r^2}{k} - \frac{T_{..}'+x^2}{rk}$
ERROR	By Subtraction
Total	$\sum_{ij} Y_{ij}^2 + x^2 - \frac{(T_{..}'+x)^2}{rk}$

Thus the error Sum of Squares is obtained as

$$SSE = x^2 + \frac{(T_{..}'+x)^2}{kr} - \frac{(T_{.1}'+x)^2}{r} - \frac{(B_1'+x)^2}{k}$$

+ terms which do not Involve x

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$$\frac{\delta(SSE)}{\delta x} = 2x + \frac{2(T_{..}' + x)}{kr} - \frac{2(T_{.1}' + x)}{r} - \frac{2(B_{1.}' + x)}{k} \quad \text{--- (1)}$$

$$2x + \frac{2(T_{..}' + x)}{kr} - \frac{2(T_{.1}' + x)}{r} - \frac{2(B_{1}' + x)}{k} = 0$$

Now Separate this terms

$$x + \frac{T_{..}'}{kr} + \frac{x}{kr} - \frac{T_{.1}'}{r} - \frac{x}{r} - \frac{B_{1}'}{k} - \frac{x}{k} = 0$$

Common x term Now

$$x\left[1 + \frac{1}{kr} - \frac{1}{r} - \frac{1}{k}\right] + \frac{T_{..}'}{kr} - \frac{T_{.1}'}{r} - \frac{B_{1.}'}{k} = 0$$

$$x\left(\frac{kr - 1 - k - r}{kr}\right) = \frac{rB_{1.}' + kT_{.1}' - T_{..}'}{kr}$$

$$x = \frac{rB_{1}' + kT_{.1}' - T_{..}'}{(k-1)(r-1)}$$

The Formula in general may be written as

$$\boxed{x = \frac{rB + kT - G}{(k-1)(r-1)}}$$ Quiz*

Where, r = num of replicates

k = number of treatment

B = Sum of remaining value in the blocks with the missing observations.

T = Sum of " " " " " the treatment with missing observations

G = Grand total of all observed values.

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Example

The following data obtained from a randomized Complete block design with 3 treatments A, B, C and 3 blocks, contain one missing observations represented by X.

Blocks	Treatments A	B	C
I	5	12	15
II	7	10	14
III	8	16	X

Estimate the missing observation and prepare a table for analysis of variance.

Sol:-

$$x = \frac{rB + kT - G}{(k-1)(r-1)}$$

$r = 3$, $k = 3$ $\quad B = 8 + 16 = 24$,

$T = 15 + 14 = 29$

$G = 87$

$$x = \frac{3(24) + 3(29) - 87}{(3-1)(3-1)} = 18$$

$$\boxed{x = 18}$$

Treatment

Blocks	A	B	C
I	5	12	15
II	7	10	14
III	8	16	18

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1 Hypothesis:-

$H_0: \mu_1 = \mu_2 = \mu_3$

H_1: Not all Treatments mean are equal

$H_0': \mu_1. = \mu_2. = \mu_3.$

H_1': NoT all blocks mean are equal

2 Level of Significance:-

$\alpha = 0.05$

3 Test Statistics

$F_1 = \frac{s_t^2}{s_e^2}$ → MS For Treatments

→ MS error

$F_2 = \frac{s_b^2}{s_e^2}$ — MS For Blocks

4 Calculation:-

$TSS = \sum_i \sum_j Y_{ij}^2 - C.F$ $\quad \because C.F = \frac{T_{..}^2}{n}$

$n = rk = 9$

$TSS = 158$ $\quad C.F = \frac{(105)^2}{9} = 1225$

$Treat\,SS = \frac{\sum T_{.j}^2}{r} - C.F = 126$

$BSS = \frac{\sum_i B_{i.}^2}{k} - C.F = 24.7$

$ESS = TSS - Treat\,SS - BSS$

$ESS = 7.3$

Putting ANOVA Table

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Source of var	Df of freedom	Sum of Sq	Mean Sq	F-Test
Treatments	$\nu_1 = k-1$ $= 3-1$ $= 2$	126 = Trt SS	$s_t^2 = \frac{126}{2}$ $= 63$	$F_1 = \frac{s_t^2}{s_e^2}$ $F_1 = 63/2.43$ $F_1 = 25.9$
Blocks	$\nu_2 = 2$	247 = BSS	$s_b^2 = \frac{BSS}{\nu_2}$ $= 12.35$	$F_2 = \frac{s_b^2}{s_e^2}$ $= 5.08$
Error	$\nu_3 = (r-1)(k-1)-1$ $\nu_3 = 3$	7.3 = ESS	$s_e^2 = 2.43$	
Total	7	158 = TSS		

5 Critical Region :-

Reject null hypothesis if $\begin{cases} F_1 \geq F_{0.05}(2,3) \\ F_2 \geq F_{0.05}(2,3) \end{cases}$

$25.9 > 9.55$ → Reject H_0

$5.08 < 9.55$ → Do not Reject H_0

6- Conclusion

As over calculated value F_1 fall in C.R so, all the treatment means are not equal, but the F_2 does n't lie in Rejection region. we conclude that all the block means are equal

Efficiency of RCBD Relative To CR Design

For experimental design, efficiency relates to the ability of a design to achieve the objective with minimal expenditure of resource such as time and money.

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Date: 1/20 Day:

2. If relative efficiency is usually expressed in terms of a ratio error variances.

* $RE(RCBD, CR) = \frac{\text{Estimated MSE for CR design}}{\text{MSE of the RCB design}}$ Our

$$RE(RCB, CR) = \frac{(k-1)S_b^2 + k(k-1)S_e^2}{(rk-1)S_e^2}$$

EXAMPLE

Given the following analysis of Variance For RCBD.

Source of Variation	Degree of freedom	Sum of Squares	Mean Squares
Blocks	9	0.4074	0.0453
Treatments	3	1.1986	0.3995
Error	27	0.6247	0.0231
Total	39	2.2307	—

Estimate the efficiency of this design relative to Completely randomized design.

D.f	M.Square
$k-1=9$	$S_b = 0.0453$
$k-1=3$	$S_t^2 = 0.3995$
27	$S_e^2 = 0.0231$
$rk-1=39$	—

$$RE = \frac{(9)(0.0453) + (10)(3)(0.0231)}{39(0.0231)}$$

$$RE = \frac{1.1007}{0.9009}$$

$$RE = 1.22 = 122\%$$

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Lecture # 5

The Latin Square Design

Description

When the variation is found in two directions, it becomes necessary to remove these two source of variation simultaneously. This is done by simultaneous blocking of experimental unit in two mutually perpendicular direction called rows and columns. Since each rows and each columns is a complete block, the grouping for a balanced arrangement is performed by imposing the restriction that each treatment must apear "once and only once" in each row and in each column.

If there are "k" treatments then the experimental area will be divide into "k" rows and "k" columns in k x k plots or experimental units. As the experiment is laid out in a square pattern and treatments are then assigned at random to plots or experimental units. Such a double blocking of experimental units and corresponding Double random assignment of treatments is called "Latin Square Design".

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Layouts

Latin Square are always constructed by rotations, e.g in the case of 4 treatments A, B, C and D, we get

A	B	C	D
B	C	D	A
C	D	A	B
D	A	B	C

A large number of distinct Latin Square can be derived by Interchanging rows and Columns of a Standard Latin Square, where a **Standard Latin Square** is a Latin Square in which treatments in the first row and first column are arranged in alphabetical or numerical order.

A Standard Latin Square is also called a reduced Latin Square.

There is only one standard Square for a 2x2 Latin Square as,

A	B
B	A

And for 3x3 Latin Square we have also one standard Square Such as,

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A	B	C
B	C	A
C	A	B

But 4X4 Latin Square, there are four Possible Standard Squares as,

A	B	C	D
B	A	D	C
C	D	A	B
D	C	B	A

A	B	C	D
B	A	D	C
C	D	B	A
D	C	A	B

A	B	C	D
B	C	D	A
C	D	A	B
D	A	B	C

A	B	C	D
B	D	A	C
C	A	D	B
D	C	B	A

Each Latin Square $k \times k$ yields $k!(k-1)!$ Different Squares by Permuting the rows except first and all the k columns. According we may obtain from a Square design.

2X2 → $2!(2-1)! = 2$

3X3 $3!(3-1)! = 12$

4X4 $4!(4-1)! = 576$

5X5 $5!(5-1)! = 161280$

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Mathematical Model

In Latin Square design, the observation $Y_{ij(h)}$,

$$Y_{ij(h)} = \mu + R_i + C_j + T_h + e_{ij(h)}$$

where

$$i, j, h = 1, 2, 3 \ldots k$$

The μ denote the general mean

R_i is the effect of the ith row

C_j the effect of the jth Column

T_h the effect of the hth treatment

$e_{ij(h)}$ is the experimental error, that is assumed to be normally and Independently distributed with mean zero and common variance σ^2.

1. Formulation of Hypothesis
2. level of Significance
3. Test Statistics
4. Calculation
5. Critical Region
6. Conclusion

STASTistical Analysis For Latin Sq Design

1 Hypothesis

$H_0: T_h = 0$

$H_1: \neq 0$

$H_0: \mu_{..1} = \mu_{..2} = \mu_{..3} \ldots$

H_1: Not all treatment means are equal

$H_0': \mu_{.1.} = \mu_{.2.} = \mu_{.3.} \ldots$ $H_0\ C_j = 0$

H_1': Not all column means are equal $H_1 \neq 0$

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H_0: $\mu_{1\cdot} = \mu_{2\cdot} = \cdots$ $H_0: R_i = 0$, $H_1: \neq 0$

H_1: Not all row mean are equal

2 Level of Significance

$\alpha = ?$

3 Test-Statistic

$F_1 = \frac{S_t^2}{S_e^2}$ — mean sq for Treatm / mean sq for error

$F_2 = \frac{S_{R}^2}{S_e^2}$ — Mean sq For row

$F_3 = \frac{S_{C}^2}{S_e^2}$ — mean sq for Colum

4 Conclusion:-

The data obtained from a $k \times k$ Latin Square Design may be arranged in tabular form as

Rows	Columns 1	2	3	4	...	k	R_i Total	Means
1	Y_{11a}	Y_{12B}	Y_{13C}	Y_{14D}	--		$R_{1\cdot\cdot}$	$\bar{Y}_{1\cdot\cdot}$
2							$R_{2\cdot\cdot}$	$\bar{Y}_{2\cdot\cdot}$
3							$R_{3\cdot\cdot}$	$\bar{Y}_{3\cdot\cdot}$
4	Y_{41D}	Y_{42A}	Y_{43B}	Y_{44C}	--			
⋮	⋮	⋮	⋮	⋮				
k								
Total C_j	$C_{1\cdot}$	$C_{2\cdot}$	--				Grand Total	
Means	$\bar{Y}_{\cdot 1\cdot}$	$\bar{Y}_{\cdot 2\cdot}$	$\bar{Y}_{\cdot 3\cdot}$	--				$\bar{Y}_{\cdot\cdot\cdot}$

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Computation For the Treatment Means

Just see handout

Partitioning of Total Sum of Squares

TSS = RSS + CSS + TreatSS + ESS

Total SS = Row SS + Column SS + Treatment SS + Error SS

k^2-1 $\quad$ $k-1$ $\quad$ $k-1$ $\quad$ $k-1$ $\quad$ $(k-1)(k-2)$

Df

$k^2-1-3(k-1)$

$k^2-1-3k+3$

k^2-3k+2

$k^2-2k-k+2$

$k(k-2)-1(k-2)$

$(k-1)(k-2)$ ⇒ Error SS

ESS = TSS − RSS − CSS − Treat SS

$$C.F = \frac{(T...)^2}{n} = \frac{G^2}{k^2}$$

$$TSS = \sum_i \sum_j (Y_{ij(h)}^2) - C.F$$

$$RSS = \frac{\sum_i R_i^2}{k} - C.F$$

$$CSS = \frac{\sum_j C_j^2}{k} - C.F$$

$$Treat\ SS = \frac{\sum_h T_h^2}{k} - C.F$$

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ANOVA Table For kxk Latin Square Design

Source of Variation	Degree of freedom	Sum of squares	Mean Square	F-Test
B/w Treatm	$\nu_1 = k-1$	Treat SS = $\sum_h T_h^2/k - C.F$	$S_t^2 = \frac{\text{Treat SS}}{\nu_1}$	$F_1 = S_t^2/S_e^2$
B/w Rows	$\nu_2 = k-1$	RSS = $\sum_i R_i^2/k - C.F$	$S_r^2 = \frac{RSS}{\nu_2}$	$F_2 = S_r^2/S_e^2$
B/w Colum	$\nu_3 = k-1$	CSS = $\sum_j C_j^2/k - C.F$	$S_c^2 = \frac{CSS}{\nu_3}$	$F_3 = \frac{S_c^2}{S_e^2}$
within Error	$\nu_4 = (k-1)(k-2)$	TSS - Treat SS - CSS - RSS = ESS	$S_e^2 = \frac{ESS}{\nu_4}$	
Total	k^2-1	$\sum_i \sum_j (Y_{ij(h)})^2 - C.F$		

5-CRitical Region :-

Reject the null hypothesis

$F_1 \geq F_{\alpha,(\nu_1,\nu_2)}$

$F_2 \geq F_{\alpha,(\nu_2,\nu_4)}$

$F_3 \geq F_{\alpha,(\nu_3,\nu_4)}$

Now EXAMPLE

The atmosphere in 4 different districts of a large town was sampled, the sample being taken at 4 different heights. Four different

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test for the presence of a certain chemical were made on the samples. The arrangement is shown in the following table with the percentage by weight of the chemical as determined by the tests. Letter denote the difference test.

Heights	Districts 1	2	3	4
1	A 8	B 5	C 4.1	D 5
2	D 6.8	A 4.9	B 4.1	3.2 C
3	B 6.3	C 4.7	D 4.0	A 5
4	C 5.7	D 3.3	A 4.0	B 4.2

Is there any evidence of significant variation from district and betwee heights in the percentage of the chemical present in the atmosphere -

1 Hypothesis ✱

$H_0: \mu_{.1} = \mu_{.2} = \mu_{.3} = \mu_{.4}$

H_1: Not all treatments means are equal

$H_0': \mu_{1.} = \mu_{2.} = \mu_{3.} = \mu_{4.}$

H_1': Not all row means are equal

$H_0'': \mu_{.1.} = \mu_{.2.} = \mu_{.3.} = \mu_{.4.}$

H_1'': Not all column means are equal

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Date: 1/20

2 Level of Significance ★

$\alpha = 0.05$

3 Test – Statistics

$F_1 = \frac{S_t^2}{S_e^2}$ — Tests

$F_2 = \frac{S_i^2}{S_e^2}$ — height

$F_3 = \frac{S_c^2}{S_e^2}$ ——→ District

4 Calculations

Height	Districts 1	2	3	4	R_i	R_i^2
1	A 8	B 5.3	C 4.1	D 5	22.4 →	501.76
2	D 6.8	A 4.9	B 4.1	C 3.2	19.0	361.00
3	B 6.3	C 4.7	D 4.0	A 5	20	400
4	C 5.7	D 3.3	A 4.0	B 4.2	17.2	295.84
C_j	26.8	18.2	16.2	17.4	Grand 78.6	1558.60
C_j^2	718.24	331.24	262.44	302.76	1614.68	
$\Sigma Y_{ij(k)}^2$	$(8)^2+(6.8)^2+(6.3)^2+(5.7)^2 = 182.42$	85.08	65.62	77.88	411.00	

Tests

	A	B	C	D	
	8	5.3	4.1	5	
	4.9	4.1	3.2	6.8	
	5	6.3	4.7	4.0	
	4.0	4.2	5.7	3.3	Total
T_h	21.9	19.9	17.7	19.1	78.6
T_h^2	479.61	396.01	313.29	364.8	1552.72

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Date: 1/120 Day:

$$TSS = \sum_i \sum_j Y_{ij(h)}^2 - C.F \qquad C.F = \frac{G^2}{k^2} = \frac{(78.6)}{(4)^2}$$

$TSS = 411.00 - 386.1225$ $\qquad C.F = 386.1225$

$TSS = 24.8775$

$$RSS = \frac{\sum R_i^2}{k} - C.F$$

$$RSS = \frac{1558.60}{4} - 386.1225$$

$RSS = 3.5275$

$$CSS = \frac{\sum C_j^2}{k} - C.F$$

$$CSS = \frac{1614.68}{4} - 386.1225$$

$CSS = 17.5475$

$$Treat\,SS = \frac{\sum T_h^2}{k} - C.F$$

$$Tss = \frac{1553.72}{4} - 386.1225$$

$Tss = 2.3075$

$ESS = T.SS - RSS - CSS - TreaSS$

$ESS = 1.4950$

Putting ANOVA Table For 4x4

Source of Variation	Degree of Freedom	Sum of squar	Mean square	F-Test
B/w Treatre	$N_1 = 4-1 = 3$	$Tss = 2.3075$	$S_t^2 = 5.8492$	$F_1 = 23.47$
B/w Rows	$N_2 = 3$	$RSS = 3.5275$	$S_j^2 = 1.1758$	$F_2 = 4.72$
B/w clm	$N_3 = 3$	$CSS = 17.5475$	$S_c^2 = 0.7692$	$F_3 = 3.09$
Error	$15-9 = 6 = N_4$	1.4950	$S_e^2 = 0.2490$	
Total	$k^2-1 = (4)^2-1 = 15$	24.8775		

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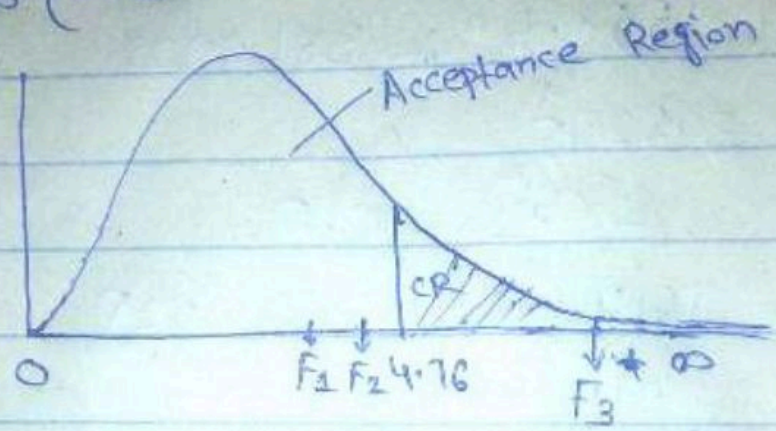
5 Critical Region

Reject Null hypothesis if

3.09 ← $F_1 > F_{0.05}(3,6) = 4.76$

4.72 ← $F_2 > F_{0.05}(3,6) = 4.76$

23.47 ← $F_3 > F_{0.05}(3,6) = 4.76$



6 Conclusion

By comparing the computed values of F with the Corresponding Critical F, we conclude that the variation among the heights is Significant. Also, there is no difference b/w the Sensitivity of the test

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